

Neal Daftary

nealdaftary0405@gmail.com • +91 91064 97430 • neal-daftary.vercel.app • neal-daftary • Neal006

Education

Nirma University, B.Tech in CSE (AI & ML) – Ahmedabad, India
• CGPA: 7.7 (up to 4th semester)

Expected 2028

Experience

Curriculo, Software Engineering Intern (Remote)

Aug 2026 – Present

- Built real-time notifications merging 5 event sources into 1 SQL query (NestJS, PostgreSQL, Next.js).
- Translated job pages into ~200 locales via NLLB and caching, cutting 30–60 DB lookups per page to 1.
- Shipped curriculo.me redesign in 8 languages on Cloudflare Pages, preserving SEO via 12 URL redirects.
- Deployed to AWS (EC2, RDS, S3) via CI/CD and closed a cross-tenant data leak in the Gmail integration.

MZHubtech, Software Engineering Intern (Remote)

Oct 2025 – Dec 2025

- Deployed a Next.js 14 SSR app on Azure App Service via GitHub Actions CI/CD & wired the frontend with Azure Cosmos DB & SendGrid SMTP for serverless contact automation.

8xSports, AI Engineering Intern (Remote)

Jun 2025 – Sep 2025

- Built a Visual Search Engine that finds visually similar jerseys in a 15,000+ image catalogue by cropping the jersey with YOLOv8 and matching DINO feature vectors, GPU-accelerated with CUDA. Earned Letter of Recommendation for high-impact execution.

Open Source Contributions

- **JAX Privacy (Google DeepMind)** – Migrated 9 modules off chex to optax.ArrayTree/jax.typing, raised the jax/jaxlib floor, and fixed CI type-checks and docs across 5 merged PRs (#301, #304, #316, #321, #333).
- **TORAX (Google DeepMind)** – Rebuilt the Plotly animation slider, cutting render time $2.5\times$ (153→67 ms) and payload size 76%, closing an issue open since 2024 (#2318).
- **Transformers (Hugging Face)** – Fixed an ImportError breaking model exporters on torch < 2.8 in a 150k+ star repo (#47710, #47711).
- **OpenCV** – Fixed 4 functions, including SIFT, to accept boolean masks by relaxing outdated type checks and removing a redundant conversion step, across 4 merged PRs (#29676, #29677, #29679, #29681).
- **Claude Code Action (Anthropic)** – Fixed 2 bugs: restored tool-result text dropped from job summaries, and fixed bot-actor filters (#1616, #1619).
- **workers-sdk (Cloudflare)** – Fixed a crash from malformed SSH config values by adding proper validation and clear errors (#15088).

Publications

IEEE Sensors Letters (SCI Q3, IF 2.2) – *CatBoost Anomaly Detection in Robotic Arms*

Jan 2026

- Reached 97.20% accuracy and 0.9718 F1-score by proposing a CatBoost anomaly detector, beating SVM, LR, NB, and QDA baselines.

Projects

Minutes | TypeScript, React, Express, SQLite, OpenRouter, Groq, Kubernetes/Argo CD

- Built an AI meeting-notes app that delivers summaries, decisions and action items 9–28 s after a meeting ends, by streaming audio to Groq Whisper and extracting notes with free OpenRouter LLMs.
- Kept AI cost at \$0, with every call confirmed in OpenRouter's own records, by adding a free-model-only guard, per-call provenance logging, fallback chains and a silent-audio filter.
- Shipped with 18 unit tests, 17 browser e2e tests and a no-mocks live test, plus a CI → Docker → Argo CD pipeline deploying to Oracle Always Free for \$0/month.

HelioOps (Team Project) | Python, FastAPI, OpenCV, LightGBM, ChromaDB RAG, WebSocket, Docker, React

- Live at helioops.dpdns.org, with the FastAPI backend and Next.js frontend each containerized via multi-stage Docker builds.
- Orchestrated on Kubernetes across staging and production, running on AWS EKS infrastructure provisioned with Terraform.
- Automated with ArgoCD GitOps and a CI pipeline that lints, tests, and builds Docker images before every deploy.

Positions of Responsibility

Nirma University ACM Student Chapter, Student Chairperson – Ahmedabad, India

Sep 2025 – Present

- Led a 150+ member tech community by launching Prompt to Prototype, a 1-day AI build challenge, and mentorship tracks.

Technical Skills

Languages: Python, C++, TypeScript, JavaScript, SQL.

Full Stack: Next.js, NestJS, Prisma, PostgreSQL, TanStack Query, Redis, BullMQ, FastAPI, Node.js, WebSocket, REST/SSE.

AI/ML: Machine Learning, Computer Vision, MLOps, Generative AI, RAG, Agentic AI, Prompt Engineering.

AI/ML Frameworks: NumPy, Pandas, SciPy, scikit-learn, PyTorch, ONNX, TensorRT, Optuna, OpenCV, Hugging Face, LangChain, LangGraph, LangSmith, AgentScope, ChromaDB, MLflow.

Cloud: AWS (S3, ECS, CloudFront, EKS), Azure (App Service, Cosmos DB), Docker, Kubernetes, Terraform, Vercel, Railway.

Tools: GitHub Actions, ArgoCD, Prometheus, Grafana, DVC, Airflow.

Achievements

- **Winner, Aubergine Track, HACKaMINeD National Hackathon (2026):** placed Top 5 nationally among 2300+ participants.
- **National Rank 4, Mitsubishi Electric Cup (6th Edition, 2026):** a national-level Factory Automation competition.